Modèles stochastiques en finance
Séance 5
Modèles à volatilité stochastique.

Ioane Muni Toke

Laboratoire MICS, Equipe FiQuant, CentraleSupélec, France

Mention Mathématiques, Modélisation et Mathématiques Financières
Parcours Finance Quantitative
Octobre 2025 - Janvier 2026

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)

- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)

- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Historical volatility, annualized volatility estimator

- Historical volatility refers to the volatility estimated over some past time period.
- In the Black & Scholes model, the log-returns over a period δ are defined by

$$r_{t,t+\delta} = \log S_{t+\delta} - \log S_t = \left(\mu - \frac{\sigma^2}{2}\right)\delta + \sigma(B_{t+\delta} - B_t),$$

  so that $\mathbf{V}[r_{t,t+\delta}] = \sigma^2\delta$.
- The standard time unit in finance (derivative product, not high-frequency) is the year. Therefore, practitionners usually compute the standard deviation of daily returns $\hat{\sigma}_{\text{daily}}$ ($\delta = 1/252$) and then compute the so-called annualized volatility estimator

$$\hat{\sigma} = \sqrt{252}\,\hat{\sigma}_{\text{daily}}.$$

- However :

Empirical distribution of log-returns is not normal

$$\hat{r}_{t,t+\delta} = \log \hat{S}_{t+\delta} - \log \hat{S}_t \neq \left(\mu - \frac{\sigma^2}{2}\right)\delta + \sigma(B_{t+\delta} - B_t)$$

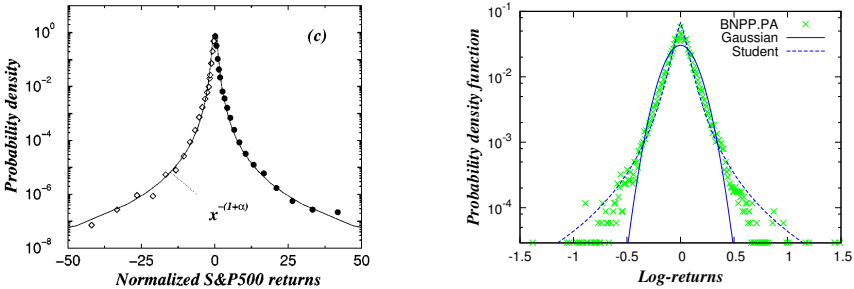


Figure: Daily normalized log-returns of the SP500 Index in 1999 (left panel) and 5-minute log-price increments for the stock BNP Paribas on Euronext from July 2007 to May 2008 (right panel), in log-log scales. Sources: see course bibliography, Gopikrishnan et al., 1999; Chakraborti et al., 2011.

Volatility is clustered

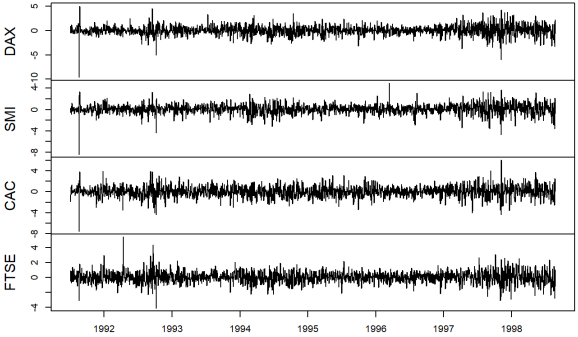


Figure: Daily log-returns for four European indices from 1991 to 1999. Extracted from `https://bookdown.org/wfoote01/faur/` on July 17th, 2018.

Implied volatility

- Let $C_{BS}(t, S_t) = C_{BS}(t, S_t; K, T; r, \sigma)$ be the Black & Scholes formula for the price at time t of the vanilla call option with strike K and maturity T.
- We have seen that the vega of a vanilla call option is $\frac{\partial C}{\partial \sigma}(t, S_t) = S_t\sqrt{T-t}\Phi'(d_1) > 0$.
- The vanilla call option price is thus an increasing function of the volatility σ with lower bound $\lim_{\sigma \to 0} C(t, S_t) = (S_t - Ke^{-r(T-t)})_+$ and uper bound $\lim_{\sigma \to +\infty} C(t, S_t) = S_t$.
- Equation $C_{BS}(t, S_t; K, T; r, \sigma) = C$ with unknown σ admits exactly one solution if C is between the above bounds.
- This solution is called the implied volatility of the vanilla call with K and maturity T. It is often written $\mathrm{IV}(K, T)$.
- On regulated markets, liquid vanilla options are often quoted in implied volatilities rather than prices.

Traders use the Black & Scholes formula... but not with a constant volatility

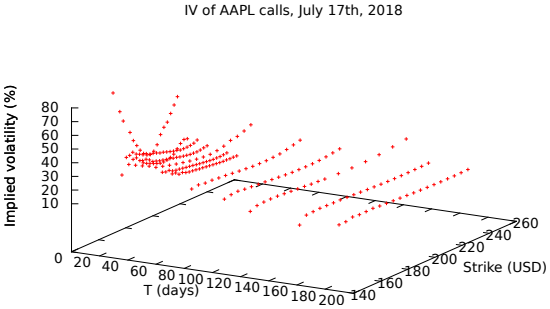


Figure: Implied volatility (IV) of quoted call options on the underlying asset Apple (AAPL) as a function of strike and maturity, on July 17th, 2018. Data retrieved on `https://www.barchart.com/quotes/AAPL/options` on July 17th, 2018.

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)
- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Model setup

- Let $T > 0$ be a finite horizon. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be some probability space with $B = (B_1, B_2)$ a two-dimensional Brownian motion with natural filtration $(\mathcal{F}_t)_{t\in[0,T]}$. Let $\rho \in]-1, 1[$. $Z_1 = B_1$ and $Z_2 = \rho B_1 + \sqrt{1-\rho^2} B_2$ are two non-independent $(\mathcal{F}_t, \mathbf{P})$-Brownian motions such that $\langle Z_1, Z_2 \rangle_t = \rho t$, $\rho \in [-1, 1]$.
- Let us consider a financial model on an risky underlying asset whose price satisfies the stochastic differential equation

$$dS_t = \mu S_t \, dt + \sqrt{v_t} S_t \, dZ_1(t),$$

for some $\mu \in \mathbb{R}$. v is called the *variance* process (and $\sqrt{v}$ is the volatility process) and we further assume that v is a $\mathbf{P}$-a.s. positive solution of the stochastic differential equation

$$dv_t = \alpha(t, S_t, v_t) \, dt + \beta(t, S_t, v_t) \, dZ_2(t).$$

- Finally, we will assume that the risk-free asset is the same as in the Black & Scholes model, with a constant risk-free rate $r > 0$.

Equivalent martingale measure in the stochastic volatility framework

Proposition 5.1 (Equivalent martingale measure in the stochastic volatility framework)

Let $\tilde{\alpha} : [0, T] \times \mathbb{R}^2 \to \mathbb{R}$ be some measurable function. With appropriate technical conditions, there exists an equivalent measure $\mathbf{Q}$ such that the stochastic volatility framework is written

$$\begin{cases} dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dZ_1^{\mathbf{Q}}(t), \\ dv_t = \tilde{\alpha}(t, S_t, v_t)\,dt + \beta(t, S_t, v_t)\,dZ_2^{\mathbf{Q}}(t), \end{cases}$$

where $Z_1^{\mathbf{Q}}$ and $Z_2^{\mathbf{Q}}$ are univariate $(\mathcal{F}_t, \mathbf{Q})$-Brownian motions such that $\langle Z_1^{\mathbf{Q}}, Z_2^{\mathbf{Q}} \rangle_t = \rho t$.

- There exists an infinite number of equivalent martingales measures.
- Practitioner's point of view : choice of $\tilde{\alpha}$ and then numerical values of parameters (and therefore martingale measure) is the role of calibration.

Pricing PDE for the general stochastic volatility framework

Proposition 5.2 (Pricing PDE for the general stochastic volatility framework)

Let $P_t = p(t, S_t, v_t)$ be the price of a European-style derivative product with maturity T in the stochastic volatility framework. $p : [0, T] \times \mathbb{R}_+^* \times \mathbb{R}_+^* \to \mathbb{R}$, $(t, x, y) \mapsto p(t, x, y)$ is a solution of the partial differential equation

$$\frac{\partial p}{\partial t}(t,x,y) + rx\frac{\partial p}{\partial x}(t,x,y) + \tilde{\alpha}(t,x,y)\frac{\partial p}{\partial y}(t,x,y) + \frac{1}{2}x^2 y\frac{\partial^2 p}{\partial x^2}(t,x,y)$$
$$+ \frac{1}{2}\beta^2(t,x,y)\frac{\partial^2 p}{\partial y^2}(t,x,y) + \rho\beta(t,x,y)x\sqrt{y}\frac{\partial^2 p}{\partial x \partial y}(t,x,y) - rp(t,x,y) = 0.$$

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)
- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Special case of an independent volatility process I

- Assume that the volatility process is completely stock-independent:

$$\begin{cases} dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dB_1(t), \\ dv_t = \alpha(t, v_t)\,dt + \beta(t, v_t)\,dB_2(t), \end{cases}$$

  where $B = (B_1, B_2)$ is a two-dimensional Brownian motion ($\rho = 0$) with respect to some risk-neutral measure $\mathbf{Q}$.
- This a strong assumption ; "usually", $\rho < 0$.
- If $\beta \equiv 0$, then v is a deterministic function of t, and the model is a time-dependent Black & Scholes model.

Special case of an independent volatility process II

- Let $\bar{v}_T$ be the mean variance along the path :

$$\bar{v}_T = \frac{1}{T}\int_0^T v_u \, du.$$

- Let $C_0^{BS}(K, T, \sigma^2)$ is the Black & Scholes price at time 0 of the call option with with strike K, maturity T and volatility σ.
- If $f_{\bar{v}_T}^{\mathbf{Q}}$ is the $\mathbf{Q}$-density of the mean variance, then the price $C_0(K, T)$ at time 0 of the standard call option with strike K and maturity T is in the above model

$$C_0(K, T) = \int_{\mathbb{R}_+} C_0^{BS}(K, T, y) \, f_{\bar{v}_T}^{\mathbf{Q}}(y) \, dy.$$

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)
- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Heston model setup

- Let $(\Omega, \mathcal{F}, \mathbf{Q})$ a probability space equipped with a filtration $(\mathcal{F}_t)_{t\in[0,T]}$. Let $\rho \in]-1,1[$ and let Z_1 and Z_2 be two univariate $\mathcal{F}_t$-Brownian motions such that $\langle Z_1, Z_2\rangle_t = \rho t$.
- The Heston models is a specification of the general stochastic volatility framework, satisfying:

$$\begin{cases} dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dZ_1(t), \\ dv_t = -\lambda(v_t - \bar{v})\,dt + \eta\sqrt{v_t}\,dZ_2(t), \end{cases}$$

  where $\eta > 0$, $\lambda > 0$ and $\bar{v} > 0$ are three positive constants.
- The variance process is known as a square-root diffusion, and specifically in finance as the CIR process, because of its proposed use to model interest rates by Cox, Ingersoll and Ross (see the Fixed Income course).
- Feller condition : solutions of square-root diffusions are $\mathbf{Q}$-a.s. positive if $2\lambda\bar{v} > \eta^2$.

Pricing with the Heston model I

- Explicit pricing: The pricing PDE of the general stochastic volatility framework becomes in the Heston case :

$$\frac{\partial p}{\partial t} + rx\frac{\partial p}{\partial S} - \lambda(y-\bar{v})\frac{\partial p}{\partial y} + \frac{1}{2}S^2 y\frac{\partial^2 p}{\partial S^2} + \frac{1}{2}\eta^2 y\frac{\partial^2 p}{\partial y^2} + \rho\eta S y\frac{\partial^2 p}{\partial S\partial y} - rp = 0.$$

An explicit solution for the call option with strike K and maturity T can be found this PDE. Sketch of the proof :

- Suitable changes of variables $\tau \leftarrow T-t$, $x \leftarrow \log\left(\frac{S}{K}e^{r\tau}\right)$ and $y \leftarrow v$ gives

$$-u_\tau + \frac{1}{2}vu_{xx} + \rho\eta v u_{xv} + \frac{1}{2}\eta^2 v u_{vv} - \frac{1}{2}vu_x - \lambda(v-\bar{v})u_v = 0.$$

- By analogy with the Black & Scholes formula, we search a solution with the form $u(\tau,x,v) = K\left[e^x P_1(\tau,x,v) - P_0(\tau,x,v)\right]$. By substitution, this gives two new PDEs for P_0 and P_1.

Pricing with the Heston model II

- These new PDEs can be solved in the Fourier domain (w.r.t. x). Taking the Fourier transform of the PDEs, we obtain and PDE on the Fourier transform $\tilde{P}_j$ of the quantity P_j. This PDE yieds solutions of the form

$$\tilde{P}_j(\tau, u, v) = \frac{1}{iu} \exp\left(C_j(\tau, u)\bar{v} + D_j(\tau, u)v\right)$$

- Finally, taking the inverse Fourier transform, one obtains:

$$P_j(\tau, x, v) = \frac{1}{2} + \frac{1}{\pi} \int_0^{+\infty} \mathrm{Re}\left\{ \frac{\exp\left[C_j(\tau, u)\bar{v} + D_j(\tau, u)v + iux\right]}{iu} \right\} du,$$

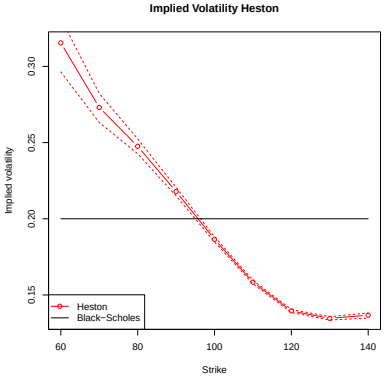
  where C and D are two independent functions of x and v.

This gives an explicit formula for the vanilla options. Pioneering work in terms of Fourier pricing. Multiple Fourier pricing formulas developped since then (see next section).

- Monte Carlo pricing: see last section and Lab

Heston smile

- Heston implied volatility smile computed via Monte Carlo with $S_0 = 100$, $r = 0.05$, $\sqrt{v_0} = \sqrt{\bar{v}} = 0.20$, $T = 1$, $\lambda = 1.0$, $\eta = 0.5$, $\rho = -0.70$



Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)
- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Fourier pricing I

Proposition 5.3 (A Fourier pricing formula)

Let us consider a general stochastic volatility model on the underlying $(S_t)_{t\in[0,T]}$ *with constant risk-free rate* r*. Let* $\mathbf{Q}$ *be an equivalent martingale measure. Let us consider a vanilla call option with maturity* T *and strike* K*. Let* $X_T = \log(S_T/S_0)$ *be the (normalized) log-price at time* T *and* $\Phi_{X_T} : u \mapsto \mathbf{E}^{\mathbf{Q}}\left[e^{iuX_T}\right]$ *its characteristic function under* $\mathbf{Q}$*. Let* $k = \log(K/S_0)$ *the log-strike. The price at time* $t = 0$ *of the call option can be written*

$$C_0 = S_0\Pi_1 - Ke^{-rT}\Pi_2$$

where

$$\Pi_1 = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathfrak{Re}\left[\frac{e^{-iuk}\Phi_{X_T}(u-i)}{iu\Phi_{X_T}(-i)}\right] du, \quad \Pi_2 = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathfrak{Re}\left[\frac{e^{-iuk}\Phi_{X_T}(u)}{iu}\right] du.$$

Fourier pricing II

- Explicit formula when the characteristic function Φ_{X_T} is known.
- Numerical integration required (e.g., Gauss-Laguerre quadrature)
- Numerous alternative formulas exist, usually derived for an easier numerical evaluation (mainly: alternative definitions of X_T, e.g., $X_T = \log\left(\frac{S_T}{K}\right) + rT$; use of properties of characteristic functions to design formulas with a single integral evaluation)
- In the Black & Scholes model,

$$\Phi_{X_T}(u) = \exp\left(iu\left(r - \frac{\sigma^2}{2}\right)T - \frac{1}{2}u^2\sigma^2 T\right).$$

In the Heston model, we have two functions A and B such that

$$\Phi_{X_T}(u) = \exp\left(A(u) + B(u)v_0\right).$$

Both are special cases of affine diffusion processes.

Fourier pricing III

Definition 5.4 (Affine diffusion)

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbf{P})$ *be a probability space and* B *be a* d*-dimensional* $\mathcal{F}_t$*-Brownian motion. Let* X *be the solution of the SDE* $dX_t = a(X_t)dt + b(X_t)dB_t$*.* X *is called an affine diffusion process if there exist* $A(t,u)$ *with values in* $\mathbb{C}$ *and* $B(t,u)$ *with values in* $\mathbb{C}^d$ *such that for all* $u \in i\mathbb{R}^d$*,*

$$\mathbf{E}[e^{\langle u, X_T \rangle} | \mathcal{F}_t] = \exp\left(A(T-t,u) + \langle B(T-t,u), X_t \rangle\right).$$

- Fourier pricing is available for financial models based on affine diffusion processes.
- A necessary condition is that there exists vectors α_i and $d \times d$-matrices β_j such that $a(x) = \alpha_0 + \sum_{i=1}^d x_i \alpha_i \in \mathbb{R}^d$ and $b^T b(x) = \beta_0 + \sum_{j=1}^d x_j \beta_j \in \mathbb{R}^d \times \in \mathbb{R}^d$.

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)
- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Stochastic volatility and implied volatility I

- Assume now that the "real world" is governed by a general stochastic volatility model:

$$\begin{cases} dS_t &= rS_t\,dt + \sqrt{v_t}S_t\,dZ_1(t), \\ dv_t &= \alpha(t, S_t, v_t)\,dt + \beta(t, S_t, v_t)\,dZ_2(t). \end{cases}$$

- Let C_{SV} be the price of the call option in this stochastic volatility model.
- Assume that a trader prices a European call option with strike K and maturity T using the Black & Scholes formula C_{BS} with some constant implied volatility $\sigma_{BS}(K, T)$.

Stochastic volatility and implied volatility II

- The difference between the two prices is:

$$C_{SV}(0, S_0) - C_{BS}(0, S_0) = \mathbf{E}^{\mathbf{Q}}\left[\int_0^T \frac{1}{2} e^{-rt} \Gamma_{BS}(t) S_t^2 \left(v_t - \sigma_{BS}^2(T, K)\right) dt\right].$$

- There is no discrepancy if

$$\sigma_{BS}^2(T, K) = \mathbf{E}^{\mathbf{Q}}\left[\int_0^T v_t \frac{e^{-rt}\Gamma_{BS}(t) S_t^2}{\mathbf{E}^{\mathbf{Q}}\left[\int_0^T e^{-rt}\Gamma_{BS}(t) S_t^2 \, dt\right]} dt\right].$$

- The implied variance in a stochastic volatility model can be seen as an expectation of the realized variance for some probability measure that weighs the realizations according to the gamma of the call option.

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)

- Discrétisation d'EDS
- Monte Carlo pour le modèle de Heston

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)

Discrétisation d'EDS

Monte Carlo pour le modèle de Heston

Principe de la discrétisation

- On considère l'équation différentielle stochastique

$$dX_t = a(X_t)dt + b(X_t)dB_t, \quad X_0 = x \ \mathbf{P}\text{-p.s.}$$

- Si on sait résoudre l'EDS, alors on peut simuler directement X_T à toute date T. Exemple : Modèle de Black & Scholes, $a(x) = rx$, $b(x) = \sigma x$,

$$X_T \stackrel{\mathcal{L}}{=} \exp\left(\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z\right)$$

  avec $Z \sim \mathcal{N}(0,1)$

- Dans le cas général, la simulation repose sur une discrétisation de l'équation, i.e. le découpage de $[0, T]$ en M pas de temps, $h = \frac{T}{M}$, et la simulation successive des approximations $\hat{X}_{jh}$ en fonction des $\hat{X}_0, \ldots, \hat{X}_{(j-1)h}$.

Erreur des schémas de discrétisation

- On appelle *erreur forte* ou *trajectorielle* du schéma de discrétisation une quantité de la forme
$$\mathbf{E}\left[\left|\hat{X}_T - X_T\right|\right] \quad \text{ou} \quad \mathbf{E}\left[\sup_{j=1,\dots,M}\left|\hat{X}_{jh} - X_{jh}\right|\right].$$
- On appelle *erreur faible* du schéma de discrétisation une quantité de la forme
$$\left|\mathbf{E}\left[f(\hat{X}_T)\right] - \mathbf{E}\left[f(X_T)\right]\right|.$$
pour toute fonction f suffisamment régulière.
- Si l'erreur forte (resp. faible) est en $\mathcal{O}(h^\delta)$, alors δ est appelé *ordre* fort (resp. faible) du schéma de discrétisation.

Schéma d'Euler-Maruyama I

- On reprend l'équation différentielle stochastique

$$dX_t = a(X_t)dt + b(X_t)dB_t, \quad X_0 = x \ \mathbf{P}\text{-p.s.}$$

- Le schéma d'Euler-Maruyama généralise le schéma d'Euler des équations différentielles ordinaires:

$$\hat{X}_{jh} = \hat{X}_{(j-1)h} + a(\hat{X}_{(j-1)h})\ h + b(\hat{X}_{(j-1)h})\ \sqrt{h}Z_j, \quad j = 1, \ldots, m,$$

  où les Z_j sont des gaussiennes centrées réduites indépendantes.
- Le schéma d'Euler est d'ordre fort $\frac{1}{2}$ et d'ordre faible 1, sous conditions de régularité sur a, b (et éventuellement f pour le cas faible).

Schéma d'Euler-Maruyama II

Theorem 5.5 (Convergence du schéma d'Euler-Maruyama)

Soit l'EDS $dX_t = a(t, X_t)dt + b(t, X_t)dB_t, \quad X_0 = x$ $\mathbf{P}$*-p.s. Si il existe des constantes* K_1, K_2 *et* K_3 *telles pour tout* $s, t \in [0, T]$,

(i) $|a(t, y) - a(t, x)| + |b(t, y) - b(t, x)| \leq K_1|y - x|$ *(Lipschitz),*

(ii) $|a(t, x)| + |b(t, x)| \leq K_2(1 + |x|)$ *(croissance linéaire),*

(iii) $|a(t, x) - a(s, x)| + |b(t, x) - b(s, x)| \leq K_3(1 + |x|)\sqrt{|t - s|}$,

alors il existe une constante C *telle que* $\mathbf{E}\left[\left|\hat{X}_T - X_T\right|\right] \leq C\, h^{-\frac{1}{2}}$

- Des conditions de régularité supplémentaires (par exemple: a, b de classe $\mathcal{C}^4$ avec des dérivées à croissance au plus polynomiale), sont requises pour énoncer que le schéma d'Euler-Maruyama scheme est d'ordre faible 1.

Dérivation des schémas de discrétisation I

- Dynamique sur $[t, t+h]$:

$$X_{t+h} - X_t = \int_t^{t+h} a(X_u)du + \int_t^{t+h} b(X_u)dB_u$$

- Terme de tendance :

$$\int_t^{t+h} a(X_u)du = \int_t^{t+h} \left[a(X_t) + \int_t^u a'(X_v)\left(a(X_v)dv + b(X_v)dB_v\right) + \frac{1}{2}\int_t^u a''(X_v)d\langle X\rangle_v \right] du$$

Dérivation des schémas de discrétisation II

- Terme brownien :

$$\int_t^{t+h} b(X_u)dB_u = \int_t^{t+h} \left[b(X_t) + \int_t^u b'(X_v)\left(a(X_v)dv + b(X_v)dB_v\right) + \frac{1}{2}\int_t^u b''(X_v)d\langle X\rangle_v \right] dB_u$$

- On observe que :

$$\int_t^{t+h}\int_t^u dvdu = \int_t^{t+h}(u-t)du = \mathcal{O}(h^2),$$
$$\int_t^{t+h}\int_t^u dB_v du = \int_t^{t+h}(B_u - B_t)du = \mathcal{O}\left(h^{\frac{3}{2}}\right),$$
$$\int_t^{t+h}\int_t^u dvdB_u = \int_t^{t+h}(u-t)dB_u = \mathcal{O}\left(h^{\frac{3}{2}}\right).$$

Dérivation des schémas de discrétisation III

puis que

$$\begin{aligned}
\int_t^{t+h} \int_t^u dB_v dB_u &= \int_t^{t+h} (B_u - B_t)\, dB_u \\
&= \int_t^{t+h} B_u dB_u - B_t (B_{t+h} - B_t) \\
&= \int_t^{t+h} \frac{1}{2} \left(d(B_u^2) - du\right) - B_t (B_{t+h} - B_t) \\
&= \frac{1}{2}(B_{t+h}^2 - B_t^2 - h) - B_t (B_{t+h} - B_t) \\
&= \frac{1}{2}(B_{t+h} - B_t)^2 - \frac{1}{2}h,
\end{aligned}$$

où on a utilisé Itô pour écrire $d(B_u^2) = 2B_u dB_u + du$.

Dérivation des schémas de discrétisation IV

- On en déduit pour le terme de tendance :

$$\int_t^{t+h} a(X_u)du = a(X_t)h + \mathcal{O}\left(h^{\frac{3}{2}}\right),$$

- et pour le terme brownien :

$$\begin{aligned}\int_t^{t+h} b(X_u)dB_u &= \int_t^{t+h} b(X_t)dB_u + \int_t^{t+h}\int_t^u b'(X_v)b(X_v)dB_v dB_u + \mathcal{O}\left(h^{\frac{3}{2}}\right)\\ &= b(X_t)\left(B_{t+h} - B_t\right) + \frac{1}{2}b(X_t)b'(X_t)\left[(B_{t+h} - B_t)^2 - h\right] + \mathcal{O}\left(h^{\frac{3}{2}}\right)\\ &\overset{\mathcal{L}}{=} b(X_t)\sqrt{h}Z + \frac{1}{2}b(X_t)b'(X_t)h\left[Z^2 - 1\right] + \mathcal{O}\left(h^{\frac{3}{2}}\right),\end{aligned}$$

avec $Z \sim \mathcal{N}(0,1)$.

Schéma de Milstein

- On considère l'EDS:

$$dX_t = a(X_t)dt + b(X_t)dB_t, \quad X_0 = x \ \mathbf{P} - p.s.$$

- Le schéma de Milstein s'écrit: $\forall j = 1, \ldots, m$,

$$\hat{X}_{jh} = \hat{X}_{(j-1)h} + a(\hat{X}_{(j-1)h})h + b(\hat{X}_{(j-1)h})\sqrt{h}Z_j + \frac{1}{2}b(\hat{X}_{(j-1)h})b'(\hat{X}_{(j-1)h})h\left(Z_j^2 - 1\right),$$

  où les Z_j sont des gaussiennes centrées réduites indépendantes.
- Le schéma de Milstein est d'ordre fort 1 et d'ordre faible 1 (sous conditions de croissance et régularité).

Programme

Empirical observations, historical volatility, implied volatility

A general stochastic volatility framework

Special case: independent volatility process

The Heston model

Fourier pricing

Stochastic volatility and implied volatility

Monte Carlo pricing with stochastic volatility models ($\longrightarrow$ TP Question V)

Discrétisation d'EDS

Monte Carlo pour le modèle de Heston

Discrétisation du modèle de Heston I

- Modèle de Heston :

$$\begin{cases} dS_t = rS_t dt + \sqrt{v_t} S_t dB_t^1, \\ dv_t = -\lambda(v_t - \bar{v})dt + \eta\sqrt{v_t} dB_t^2, \end{cases}$$

  avec $\langle B_1, B_2 \rangle_t = \rho t$, $\rho \in]-1, 1[$.

- Avec les prix logarithmiques $X_t = \log(S_t)$:

$$\begin{cases} dX_t = \left(r - \frac{v_t}{2}\right) dt + \sqrt{v_t} dB_t^1, \\ dv_t = -\lambda(v_t - \bar{v})dt + \eta\sqrt{v_t} dB_t^2. \end{cases}$$

- Le processus v est presque sûrement strictement positif s'il vérifie la condition de Feller: $2\lambda\bar{v} > \eta^2$ (qui n'est pas empiriquement vérifiée...)

Discrétisation du modèle de Heston II

- Discrétisation par un schéma d'Euler :

$$\begin{cases} \hat{X}_{t+\Delta t} = \hat{X}_t + \left(r - \frac{\hat{v}_t}{2}\right)\Delta t + \sqrt{\hat{v}_t}\sqrt{\Delta t}Z_1 \\ \hat{v}_{t+\Delta t} = \hat{v}_t - \lambda(\hat{v}_t - \bar{v})\Delta t + \eta\sqrt{\hat{v}_t}\sqrt{\Delta t}Z_2 \end{cases}$$

  avec Z_1 et Z_2 gausiennes centrées réduites de corrélation ρ.
- Le processus $\hat{v}$ peut prendre des valeurs négatives avec une probabilité non nulle $\mathbf{P}(\hat{v}_{t+\Delta t} < 0) = \Phi\left(\frac{\lambda(\hat{v}_t - \bar{v}) - \hat{v}_t}{\eta\sqrt{\hat{v}_t}\sqrt{\Delta t}}\right)$
- Numériquement, il faut imposer des conditions au bord, par exemple absorbantes ($\hat{v}_t \longrightarrow \hat{v}_t^+$), réfléchissantes ($\hat{v}_t \longrightarrow |\hat{v}_t|$), mixtes...

TP Question V

1. Discrétiser l'EDS de Black & Scholes avec un schéma d'Euler et un schéma de Milstein. Illustrer les convergences et retrouver les ordres forts et faibles de chacun des schémas.
2. Implémenter le pricing d'un call standard dans le modèle de Heston par Monte Carlo, avec le schéma d'Euler simple proposé.
3. Implémenter le calcul de la la volatilité implicite de Black & Scholes par dichotomie. On peut se ramener à un intervalle borné en écrivant $S_t = \frac{x}{1-x}$, $x \in [0, 1]$.
4. Tracer un exemple de smile de volatilité dans le modèle de Heston. Commenter.
5. Montrer la déformation du smile de volatilité en fonction des paramètres de la dynamique de volatilité λ, η et ρ. Commenter.
6. *(Pour aller plus loin)* Reprendre les questions deux dernières questions en ajoutant aux graphes les smiles obtenus à partir du prix calculé par la méthode de Fourier (pensez à toujours citer vos sources le cas échéant). Commenter.